BRANDEIS MASTERS IN FINANCE | 2026

SSGA QUANTITATIVE RESEARCH PROJECT

Meta-Labeling:

Teaching a Model When to Trust Itself

This report presents a quantitative investment research project exploring whether a machine learning filter can improve upon a rules-based macro trading model. We explain every concept from first principles, with technical depth available for quantitative readers.

THE PRIMARY MODEL

M1: When to BUY or SELL

A rules-based macro signal using 6 market indicators with dynamic weights.

THE SECONDARY MODEL

M2: Should We Trust M1?

A machine learning filter that predicts when M1's signal is likely to succeed.

THE RESULT

An Honest Finding

M2 did not add value in this iteration. We explain why and what comes next.

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SECTION 1

THE PROBLEM

What gap in investing are we addressing?

Most investment models tell you WHAT to buy or sell. They rarely tell you HOW CONFIDENT to be in each individual signal. This section explains why that gap matters and how meta-labeling is designed to fill it.

The Core Problem: Not All Signals Are Equal

Why even a good model sometimes gets things wrong

Imagine a weather forecast

A weather app tells you: 'It will rain tomorrow.' You trust that. But what if the app also told you its confidence level? '70% chance of rain in summer, but only 45% confidence in spring because spring weather is unpredictable.' That extra layer of information is exactly what this project builds.

The investing equivalent

Our primary model (M1) looks at financial market data and says BUY or SELL each month. It is right about 70% of the time over 28 years - a strong result. But that means it is wrong 30% of the time. The question is: can we predict WHICH 30% is likely to be wrong - before it is wrong?

What meta-labeling tries to do

Meta-labeling adds a second layer. After M1 says BUY or SELL, a second model (M2) asks: 'Given everything I know about current market conditions, should I trust this signal?' If M2 says yes, we trade. If M2 says no, we sit in cash.

The Two-Model Architecture

```
MARKET DATA  -->  M1 (Primary)  -->  BUY or SELL signal
                                          |
                                     M2 (Secondary)
                                YES: execute trade
                                NO:  sit in cash
```

Why this matters for investors

A model that trades less but only when it has high conviction can achieve better risk-adjusted returns than one that acts on every signal. Fewer but higher-quality trades means lower transaction costs, lower drawdowns, and more consistent performance.

SECTION 2

THE DATA

28 years of monthly market history

Good models require good data. This section explains what market data we use, why these specific assets were chosen, and how 28 years of history gives us a picture of how markets behave across full economic cycles including recessions, crashes, and recoveries.

What Data Do We Use?

Four asset classes, monthly frequency, 1996 to 2025

The four market building blocks

Every month from 1996 to 2025 we observe four key financial instruments. These were chosen because together they capture most of what drives investment returns across economic cycles:

S&P 500 (SPX)

The US stock market. Tracks the 500 largest US companies. When the economy is growing, stocks rise. When investors are fearful, stocks fall. This is our primary gauge of equity market health.

For technical readers: Technical: price index used for trend calculation. Monthly total returns used for performance attribution.

Bloomberg Commodity Index (BCOM)

A basket of raw materials: oil, gold, wheat, copper, etc. Commodities tend to rise when the economy is strong and inflation is picking up. They fall when demand slows.

For technical readers: Technical: price index for trend; return series for breadth calculation. Provides non-equity economic momentum signal.

US Corporate Bonds

Bonds issued by companies. They pay more than government bonds because they carry some risk. When companies are doing well, corporate bonds are in demand. When credit stress rises, they underperform government bonds.

For technical readers: Technical: monthly total returns. Used in credit vs rates spread calculation to measure risk appetite in fixed income.

US 10-Year Treasury Bond

The safest investment - the US government's 10-year borrowing. When investors are scared, they flee to Treasuries (prices rise, yields fall). Used as the risk-free benchmark in our model.

For technical readers: Technical: price index for yield momentum; returns as risk-free benchmark. Duration adjustment applied for total return calculation.

Why monthly frequency?

M1 is designed as a strategic allocation model, not a day-trading system. Monthly signals reduce noise from short-term price fluctuations and align with how institutional investors rebalance portfolios. 28 years gives us 336 monthly observations covering 4 full market cycles: dot-com boom/bust, 2008 crisis, COVID crash, and the post-2022 rate hiking cycle.

SECTION 3

M1: THE PRIMARY MODEL

How we generate BUY and SELL signals

M1 is the foundation of the system. It watches 6 financial market indicators, converts them into a single combined score, and decides each month whether to go risk-on (BUY) or risk-off (SELL). Every step uses only data that would have been available at that point in history - no future information is ever used.

3a - The 6 Market Indicators

What each one measures and why it belongs in the model

What is an indicator?

An indicator is a number calculated from market data that tells us something useful about the current state of the economy or financial markets. Think of them as six different gauges on a car dashboard - each measuring something different, and together giving a full picture of the engine.

1. SPX Trend - Is the stock market in an uptrend or downtrend?

Compares the current S&P 500 price to its average over the last 6 months. A positive number means the market is above its recent average - momentum is up. A negative

```
spx_price / rolling_mean_6m - 1.0
```

2. BCOM Trend - Is the commodity market trending up or down?

Same calculation as SPX Trend but applied to commodities. Rising commodity prices generally signal strong economic demand and inflationary pressure.

```
bcom_price / rolling_mean_6m - 1.0
```

3. Credit vs Rates - Are corporate bonds beating government bonds?

Takes the average return of corporate bonds over the last 3 months and subtracts the average return of government bonds. A positive value means investors are comfortable

```
mean(corp_ret, 3m) - mean(treasury_ret, 3m)
```

4. Risk Breadth - How broad is the risk-on sentiment across all assets?

Averages the returns of stocks, commodities, and corporate bonds, then subtracts treasury returns. When multiple asset classes are outperforming safe bonds simultaneously, market

```
mean(spx + bcom + corp, 3m) - mean(treasury, 3m)
```

5. BCOM Acceleration - Is commodity momentum speeding up or slowing down?

Compares the short-term commodity trend (3-month average return) to the longer-term trend (12-month average return). Positive means momentum is accelerating - early signal of

```
bcom_ret.rolling(3m).mean() - bcom_ret.rolling(12m).mean()
```

6. Yield Momentum - Which way are interest rates moving?

Tracks the direction of 10-year treasury prices (which move opposite to yields). A falling treasury price means rising rates - often a sign of economic strength or inflationary pressure.

```
-(treasury_price.diff(3))
```

3b - Z-Scores: A Universal Language

How we make 6 different indicators comparable to each other

The problem with raw numbers

Each of our 6 indicators uses completely different units and scales. SPX Trend might be 0.04 (4% above average), while Credit vs Rates might be 0.002 (0.2% per month difference). How do you add these together meaningfully? You cannot - unless you convert them to a common language first.

The intuition: how unusual is this reading?

Imagine you are told today's temperature is 25 degrees. Is that unusual? It depends on WHERE you are and WHAT TIME OF YEAR it is. In Siberia in January, that is extreme. In Miami in July, it is normal. Context is everything.

A Z-score answers: 'How unusual is today's indicator reading compared to its own history?' It measures the distance from the historical average in units of standard deviation. This makes all 6 indicators directly comparable.

The Z-Score Formula - Explained Simply

Z-score = (Today's value - Historical average) / Historical standard deviation

Z = 0.0 means perfectly normal. Z = +2.0 means unusually strong. Z = -2.0 means unusually weak.

A concrete example

Say the SPX Trend over 28 years has averaged +0.5% with a standard deviation of 2.0%. Today it reads +4.5%. The Z-score is (4.5 - 0.5) / 2.0 = +2.0. This tells us today's equity trend is 2 standard deviations above normal - a strong bullish signal. A Z-score of -1.5 on Credit vs Rates would mean credit is underperforming bonds by 1.5 standard deviations - a warning sign.

The critical detail: expanding history

A naive Z-score would use the full 28-year average to evaluate every single month - including future months the model hadn't seen yet. That would be cheating. Our implementation uses an EXPANDING window: the Z-score in March 2005 is computed using only data from 1996 to February 2005. The model only knows what was knowable at that moment in time.

CODE: Expanding Z-Score (src/metalabel/primary/signals.py)

```
def expanding_zscore(series, min_periods=12):
    x        = series                          # today's indicator value
    history  = x.shift(1)                      # only use PAST data (shift by 1)
    mean     = history.expanding(min_periods).mean()   # growing historical avg
    std      = history.expanding(min_periods).std()    # growing historical std
    z        = (x - mean) / std                # how unusual is today?
    return z

# The shift(1) is critical: it ensures today's value is never used
# to compute the mean/std against which today is evaluated.
```

3c - Dynamic Weights: How M1 Decides Which Indicators to Trust

Spearman Information Coefficient and inverse-correlation weighting explained

The problem with fixed weights

The simplest approach would be to weight all 6 indicators equally - just average their Z-scores. But this ignores two important problems: first, some indicators may be more predictive than others in the current market environment. Second, some indicators may be telling you the same thing as others, so you are effectively double-counting that signal.

Problem 1: Redundancy - Are two indicators saying the same thing?

If SPX Trend and Risk Breadth both rise and fall together, counting both equally gives too much weight to that one underlying factor. Our model penalises indicators that are highly correlated with each other.

The solution is called inverse-correlation weighting. Think of it like this: imagine 5 friends voting on where to go for dinner. If 4 of them always vote the same way and 1 always votes differently, the 1 dissenter has more independent information. We give more weight to the 'independent dissenter' indicators and less to the 'groupthink' indicators.

Problem 2: Relevance - Has this indicator been useful recently?

This is where the Spearman Information Coefficient (IC) comes in. Once we have 36 months of history, we look back at the last 3 years and ask for each indicator: did high readings in that indicator tend to coincide with good returns? If yes, keep trusting it. If no - zero it out.

What is Spearman Correlation?

Spearman correlation ranks things from lowest to highest, then asks: when the indicator was ranked HIGH, was the return also ranked HIGH? It is robust to outliers and does not assume a straight-line relationship - it just asks 'do they move in the same direction?'

A simple example of Spearman IC in action

Suppose over the last 3 years, every time Credit vs Rates was in its top half of readings, the next month's portfolio return was also in its top half. Spearman IC would be close to +1.0: strong positive relationship. Now suppose SPX Trend had no consistent relationship - sometimes high readings led to good returns, sometimes not. Spearman IC near 0. The model would downweight SPX Trend and upweight Credit vs Rates.

CODE: Dynamic Weighting with Spearman IC (src/metalabel/primary/signals.py)

```
# Step 1: compute inverse-correlation weights (reduce redundancy)
corr_matrix   = zscore_window.corr()          # how correlated are indicators?
sum_corr      = corr_matrix.sum(axis=0)       # total correlation each has
inv_corr      = 1.0 / sum_corr                # invert: less correlated = more weight
base_weights  = inv_corr / inv_corr.sum()     # normalise to sum to 1

# Step 2: Spearman IC filter (has this indicator predicted returns recently?)
if months_of_history >= 36:
    ic_window = last_36_months_of_data
    corrs     = ic_window.corr(method='spearman')['target_return']
    ic_mask   = (corrs > 0.0).astype(float)   # 1 if useful, 0 if misleading

# Step 3: combine - zero out misleading indicators
final_weights = base_weights * ic_mask
```

```
final_weights = final_weights / final_weights.sum()  # re-normalise
```

Why this makes M1 'dynamic'

These weights are recalculated fresh every single month. In a period where equity trends are not predicting returns well (e.g. a liquidity-driven rally), SPX Trend gets zeroed out and credit indicators get upweighted. The model continuously rewires itself based on what has been working in the recent past.

3d - From Score to Signal: The BUY / SELL Decision

How 6 numbers become one trading decision

The composite score

Once we have Z-scores for all 6 indicators and their dynamic weights, we compute a single weighted average called the composite score. This is one number that summarises the entire macro environment as the model sees it.

A high positive composite score means most indicators are signalling risk-on conditions - equities trending up, credit outperforming, broad momentum positive. A large negative score means the opposite - defensive positioning is warranted.

The Signal Decision Rule

`Composite Score > +0.31`	**BUY**	Go risk-on: overweight equities and commodities
`Composite Score < -0.31`	**SELL**	Go risk-off: overweight treasuries defensively
`-0.31 to +0.31`	**HOLD**	No clear signal: maintain previous allocation

What happens when a signal fires?

When M1 says BUY, the portfolio rotates into equities and commodities in equal proportion. When it says SELL, it rotates defensively into government bonds. The model holds its position until the composite score crosses the threshold in the opposite direction. HOLD months maintain the previous allocation unchanged.

Over 28 years, M1 generated 138 actionable signals: 66 BUY signals and 72 SELL signals. These are the events on which M2 is trained and evaluated.

Why +/- 0.31?

The threshold was calibrated on the training data to produce a reasonable signal frequency. Too tight a threshold generates too many signals (high turnover, high costs). Too wide a threshold misses meaningful moves. The 0.31 value balanced these competing objectives across the full historical sample.

3e - M1 Performance Results

28-year backtest vs four benchmarks, net of transaction costs

How does M1 compare to simply holding the market?

The ultimate test of any active model is whether it outperforms passive alternatives. We compared M1 against four benchmarks over the full 28-year period from 1996 to 2025:

Buy and Hold S&P 500	Simply hold US equities forever.	*The classic passive equity benchmark.*
60/40 Portfolio	60% equities, 40% bonds, rebalanced monthly.	*The standard institutional benchmark.*
Equal Weight Portfolio	Equal allocation across all 4 assets.	*Naive diversification baseline.*
Simple Trend	A naive momentum strategy without dynamic weights.	*Tests whether complexity adds value over s*

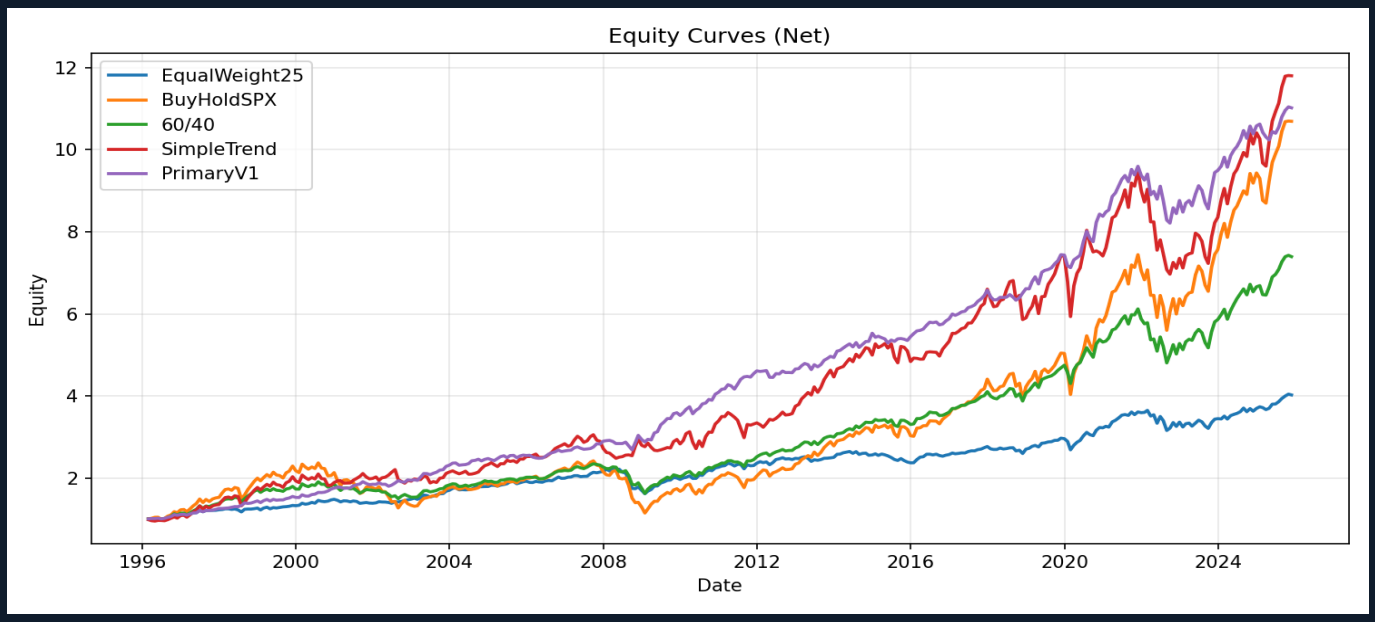


Figure 1: Cumulative equity curves 1996-2025, net of transaction costs. PrimaryV1 (purple) is our M1 model.

Key performance metrics (M1 - PrimaryV1)

Dataset period	**1996 - 2025 (28 years, monthly)**
Total M1 signals	**138 (66 BUY, 72 SELL)**
Out-of-sample win rate (actionable signals)	**71.8%**
Average monthly return per signal	**+0.76%**
Annualised Sharpe Ratio (OOS)	**1.31**
M1 vs Buy-and-Hold S&P 500	**Competitive returns with lower drawdowns**
M1 vs 60/40	**Higher Sharpe through dynamic allocation**

What the equity curve tells us

M1 (purple line) tracks closely with the best-performing benchmarks throughout history, but critically preserves capital during major drawdowns (2008, 2020) where its SELL signals moved the portfolio to safety. This defensiveness is visible in the chart as shallower dips during crisis periods.

SECTION 4

M2: THE META-LABELING FILTER

Teaching the model when to trust itself

M1 is a strong model but it is not perfect. M2 is a machine learning layer that sits on top of M1 and asks: given today's market conditions, how confident should we be in M1's signal? This section explains the concept, the construction, and - honestly - the results.

4a - What is Meta-Labeling?

A concept from quantitative finance by Marcos Lopez de Prado

The sports analogy

Imagine a football coach (M1) who calls plays based on studying the opponent. The coach is good - right about 70% of the time. Now imagine an assistant coach (M2) who watches film and says: 'Coach, I notice you tend to get this wrong in wet weather against fast defenses - maybe sit this one out.' M2 does not override the coach. It just says when to trust the coach and when to hold back.

The technical concept

Meta-labeling was introduced by Dr. Marcos Lopez de Prado in his book 'Advances in Financial Machine Learning' (2018). The key insight is:

- M1 decides the DIRECTION of the trade (BUY or SELL)
- M2 decides the SIZE or CONFIDENCE of that trade (trade vs do not trade)
- M2 is only trained on the events where M1 already fired a signal
- M2's target is binary: did M1's signal turn out to be profitable? (1=yes, 0=no)

Why not just improve M1 directly?

Because M1 uses one type of information (macro momentum trends) and M2 can use completely different information (market regime, fear levels, M1's own recent track record). These are complementary, not competing. Meta-labeling preserves M1's directional logic while adding a regime-aware confidence filter on top.

The Meta-Label Target Variable

For each of the 138 M1 signals, we look at what actually happened next month. If M1 said BUY and the portfolio made money: meta_label = 1. If M1 said BUY and the portfolio lost money: meta_label = 0. M2's job is to predict this 1 or 0 before the trade is made.

4b - New Features: VIX and OAS

Market fear and credit stress as regime context for M2

Why add VIX and OAS?

M1 uses macro trend indicators to decide direction. But it has no awareness of the broader market regime - specifically, how fearful or stressed the market is right now. Our hypothesis was: maybe M1's signals are more reliable in calm markets than in panicking ones.

VIX - The Fear Index

VIX is the CBOE Volatility Index. It measures how much volatility options traders expect in the S&P 500 over the next 30 days. Think of it as a thermometer for market anxiety. VIX of 12 = calm. VIX of 40 = panic (as seen in March 2020, October 2008).

Our hypothesis: when VIX is very high, markets are chaotic and trend signals break down. M2 should learn to be more cautious with M1's signals during high-fear regimes.

OAS - The Credit Stress Gauge

OAS (Option-Adjusted Spread) is the extra yield investors demand to hold corporate bonds instead of risk-free government bonds. Wide spreads mean investors fear corporate defaults - a sign of financial stress and illiquidity. Tight spreads mean credit markets are functioning normally.

Our hypothesis: wide and widening credit spreads signal that the financial plumbing is under stress. In such conditions, M1's macro signals may be less reliable because dislocations overwhelm fundamentals.

The theoretical link to M1

M1 already captures some of this information indirectly: credit vs rates spread and risk breadth both respond to market stress. This turned out to be an important clue about why M2 struggled - more on this in Section 5.

What we expected M2 to learn

High VIX + VIX Rising	Markets panicking. M1 trend signals unreliable. Skip trade.
Low VIX + VIX Falling	Markets calm. M1 operating in favourable conditions. Take trade.
Wide OAS + OAS Widening	Credit stress mounting. M1 SELL signals more likely correct.
Tight OAS + OAS Falling	Credit normalising. BUY signals more trustworthy.

4c - Feature Engineering

Turning raw VIX and OAS numbers into 10 model-ready inputs

Why not just use the raw VIX number?

A raw reading of VIX = 28 is not interpretable on its own. Is 28 high? It depends on history. In 1993, it would be extraordinary. In 2020, it was moderate. The model needs context. We apply the same Z-score logic as M1, plus additional features that capture different dimensions of the regime.

VIX Features (5 inputs)

`vix_level_z`	How high is VIX relative to its own full history? (expanding Z-score)
`vix_change_z`	How much did VIX move this month, vs. typical monthly moves?
`vix_trend`	Is VIX above or below its own 6-month rolling average?
`vix_high_regime`	BINARY: is VIX above its all-time expanding median? (1=yes, 0=no)
`vix_rising`	BINARY: did VIX increase this month? (1=yes, 0=no)

OAS Features (5 inputs)

`oas_level_z`	How wide are spreads relative to their full history?
`oas_change_z`	How much did OAS move this month vs. typical moves?
`oas_trend`	Is OAS above or below its own 6-month rolling average?
`oas_wide_regime`	BINARY: are spreads in a historically wide/stressed regime?
`oas_widening`	BINARY: did spreads widen this month? (credit getting worse?)

CODE: Building VIX Features (src/metalabel/secondary/features.py)

```
def build_vix_features(vix, trend_window=6, zscore_min_periods=12):
    # Level: how high is VIX vs its own history?
    level_z      = expanding_zscore(vix)              # same no-lookahead zscore

    # Change: how much did VIX move this month?
    change       = vix.diff()                         # month-over-month change
    change_z     = expanding_zscore(change)           # z-score of that change

    # Trend: is VIX above its recent average?
    rolling_mean = vix.rolling(trend_window).mean()
    trend        = (vix / rolling_mean) - 1.0         # % deviation from avg

    # Regime flags: binary 0/1 signals
    high_regime  = (vix > expanding_median(vix)).astype(float)
    rising       = (change > 0).astype(float)         # did VIX go up?

    return [level_z, change_z, trend, high_regime, rising]
```

4d - Walk-Forward Training: No Peeking at the Future

How we ensure M2 is evaluated fairly on data it has never seen

The overfitting problem in finance

Any model can be made to look perfect if it is trained and tested on the same data. Like studying with the exact exam questions - you would score 100% but learn nothing. In finance this is called overfitting, and it is the most common way models fail when deployed.

The walk-forward solution

We use an expanding walk-forward approach. Imagine you are back in January 2005. You train M2 on every signal from 1996 to December 2004 (60 months). Then you predict February 2005. Then add February 2005 to your training set and predict March 2005. And so on - never looking at future data.

Walk-Forward Illustration (simplified)

```
Step 1:  TRAIN on events 1-60    -->  PREDICT event 61
Step 2:  TRAIN on events 1-61    -->  PREDICT event 62
Step 3:  TRAIN on events 1-62    -->  PREDICT event 63
  ...    (window expands by 1 each step)
Step 78: TRAIN on events 1-137   -->  PREDICT event 138

Result: 78 out-of-sample predictions. M2 never saw any of these
events during training. This is a genuine simulation of live use.
```

Why 60 months minimum training?

Logistic regression needs enough examples to learn meaningful patterns. With only 12 training events, the model would be fitting noise. 60 months (5 years) gives a statistically reasonable sample while still leaving 78 months of genuine out-of-sample evaluation - enough to draw conclusions.

The result: 78 genuinely out-of-sample predictions

The 78 OOS events span 2009 to 2025 - a period that includes the European debt crisis, China slowdown, COVID crash, rate hiking cycle, and the AI boom. M2 was evaluated across a wide range of market conditions it had never trained on.

4e - Logistic Regression: How M2 Makes Its Decision

The machine learning model and why we chose it for this problem

What M2 is doing

M2 takes 22 input features (the 6 M1 indicators + 10 VIX/OAS features + 4 M1 track record features + M1 direction) and outputs a single number: the probability that M1's next signal will be profitable.

Why Logistic Regression?

Logistic regression is the standard starting point for binary classification in quantitative finance. It takes all the input features, multiplies each by a learned weight, sums them up, and squashes the result into a probability between 0 and 1 using the sigmoid function. Simple, interpretable, and it produces well-calibrated probabilities.

Small dataset (138 events)

Complex models like neural networks or gradient boosting need thousands of examples to avoid overfitting. With 138 events, logistic regression's built-in regularisation (the C parameter) is essential.

Calibrated probabilities

The threshold gate needs real probabilities, not just rankings. Logistic regression produces well-calibrated probabilities out of the box. Tree-based models require additional calibration steps.

Interpretability

At this research stage, we want to understand what the model is learning. Logistic regression coefficients directly show which features drive decisions.

Research discipline

Starting simple is standard practice. You establish a baseline with logistic regression before justifying the added complexity of ensemble methods.

CODE: M2 Model Configuration (src/metalabel/secondary/model.py)

```
def _make_model():
    return LogisticRegression(
        C             = 0.5,          # regularisation: prevents overfitting
        max_iter      = 500,          # iterations for solver convergence
        solver        = 'lbfgs',      # efficient for small datasets
        class_weight  = 'balanced',   # adjusts for 70/30 win/loss imbalance
        random_state  = 42,           # reproducibility
    )

# At each walk-forward step:
scaler      = StandardScaler()        # normalise features (mean=0, std=1)
X_train_s   = scaler.fit_transform(X_train)   # fit scaler on train only
X_test_s    = scaler.transform(X_test)         # apply same scale to test
model.fit(X_train_s, y_train)         # learn weights
probs       = model.predict_proba(X_test_s)[:,1]  # P(M1 wins)
m2_approve  = (probs >= 0.5).astype(int)       # threshold gate
```

4f - M2 Results: What the Data Showed

An honest assessment of the first iteration

The headline numbers

Out-of-sample events	**78 (spanning 2009-2025)**
Minimum training window	**60 months (5 years)**
Threshold used	**0.5 (default)**
M2 classification accuracy	**53.9% (random guessing = 50%)**
M2 ROC-AUC	**0.523 (random coin flip = 0.500)**

Economic performance comparison

Group	N	Win Rate	Avg Return	Ann. Sharpe
M1 baseline (all 78 OOS trades)	78	71.8%	+0.76%	1.31
M2 Approved (trade these)	46	71.7%	+0.75%	1.18
M2 Rejected (sit in cash)	32	71.9%	+0.78%	n/a

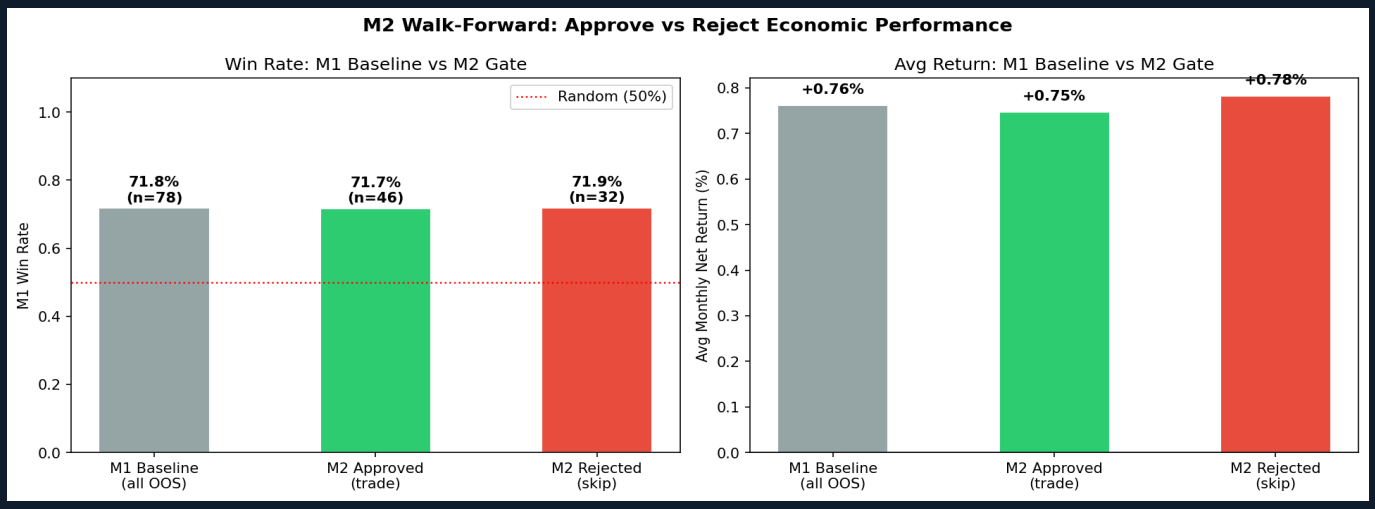


Figure 2: M2 Approve vs Reject economic performance comparison. All three groups show near-identical win rates - M2 has no discriminating power.

The honest conclusion

M2 did not add value in this iteration. The trades it approved performed slightly WORSE than the ones it rejected. The Sharpe ratio fell from 1.31 to 1.18 when filtering through M2. The model is not distinguishing winners from losers.

4f continued - Understanding Why M2 Struggled

Three hypotheses for the null result

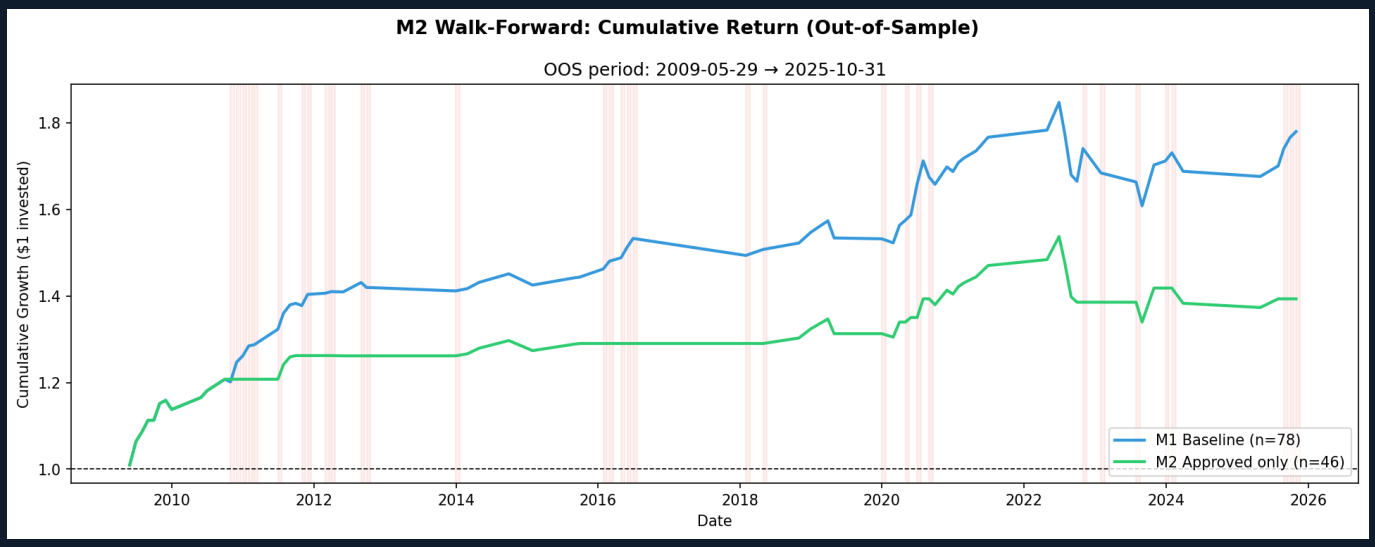


Figure 3: Cumulative return - M1 baseline (blue) vs M2-approved only (green). M2 filtering consistently underperforms full M1 across the OOS period.

Hypothesis 1: M1 already absorbs the stress signal

M1's own indicators - credit vs rates, risk breadth, SPX trend - all respond to market stress. When VIX is high, these indicators have already adjusted. By adding VIX and OAS, we may be giving M2 redundant information that M1 already processed. This is called multicollinearity: two features measuring the same underlying thing.

For technical readers: This is the most likely explanation. VIX and OAS are consequences of the same market stress that M1's indicators already capture - they add no independent signal.

Hypothesis 2: M1 is robust across all regimes by design

M1 has both BUY and SELL signals. In a market crash (high VIX), M1 is likely already positioned SELL - and that SELL is correct. M1's win rate of ~72% holds across calm and stressed markets alike, meaning there is no regime where M1 predictably fails - so no regime filter can help.

For technical readers: The data supports this: high VIX + wide OAS actually showed the BEST M1 performance (72.7% win rate, +1.02% avg return). The 'worst' regime was the best.

Hypothesis 3: 138 events is statistically too small

Even if a real pattern exists, detecting it requires enough data. With 138 monthly observations across 28 years, we have very few genuine crisis periods (2008, 2011, 2020). Statistical tests simply cannot confirm a pattern that appears in only a handful of observations.

For technical readers: The chi-square p-values of 0.89 (VIX) and 0.77 (OAS) reflect this uncertainty. We need either more events or stronger features to get below the 0.05 threshold.

SECTION 5

EDA: TESTING OUR ASSUMPTIONS

Statistical tests before building the model

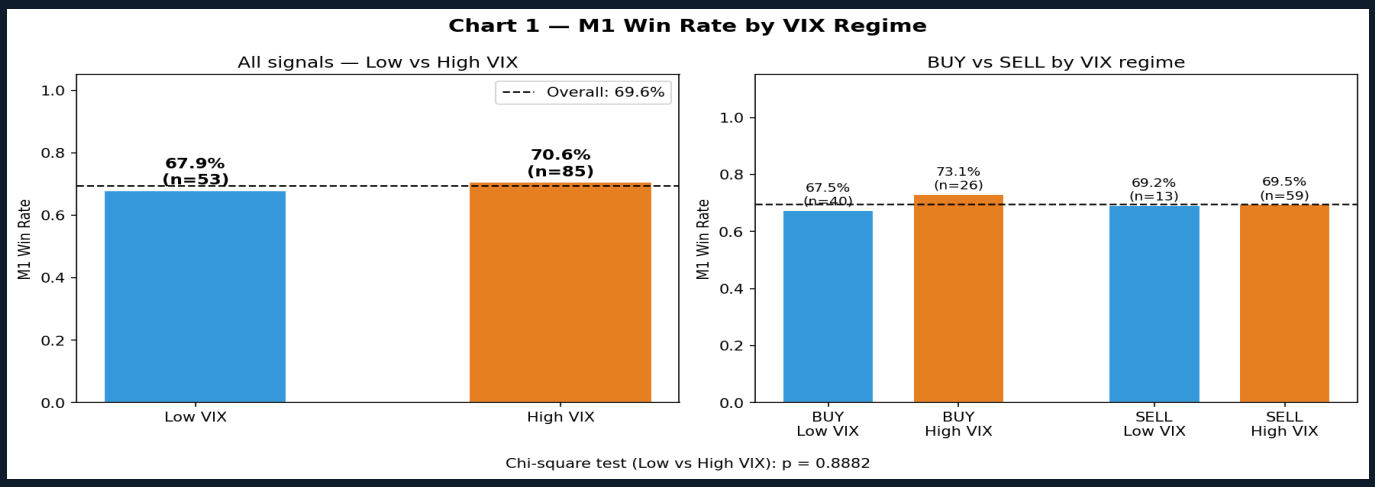
Before training M2, we conducted Exploratory Data Analysis (EDA) to test whether VIX and OAS even have a measurable relationship with M1's performance. This section presents 4 of the 6 key charts and their statistical conclusions.

5 - What the EDA Charts Showed

Six charts, one consistent finding

Chart 1 - Does VIX predict M1 wins?

We split all 138 M1 events into Low VIX and High VIX regimes and compared win rates. A chi-square test measured whether any observed difference was statistically real or just random variation with our small sample.



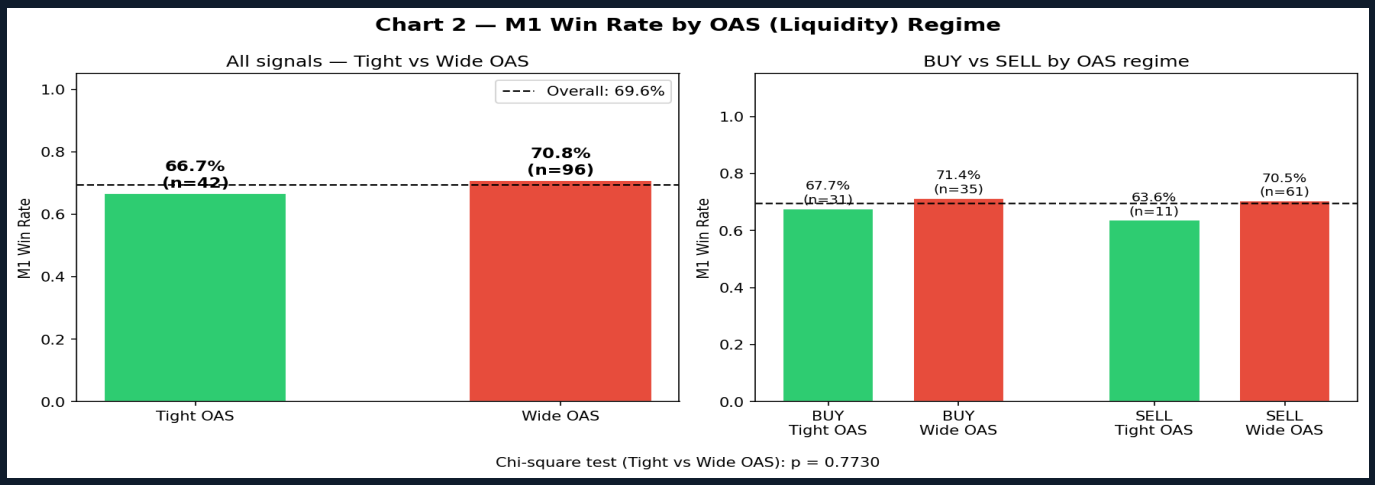
Low VIX: 67.9% win rate | High VIX: 70.6% win rate | Chi-square p = 0.8882 (not significant)

Interpretation

p = 0.8882 means there is an 89% chance this 2.7% difference is pure random noise. We need p < 0.05 to call something real. VIX regime has zero statistically significant relationship with M1 winning.

5 continued - More EDA Findings

Chart 2 - Does OAS predict M1 wins?

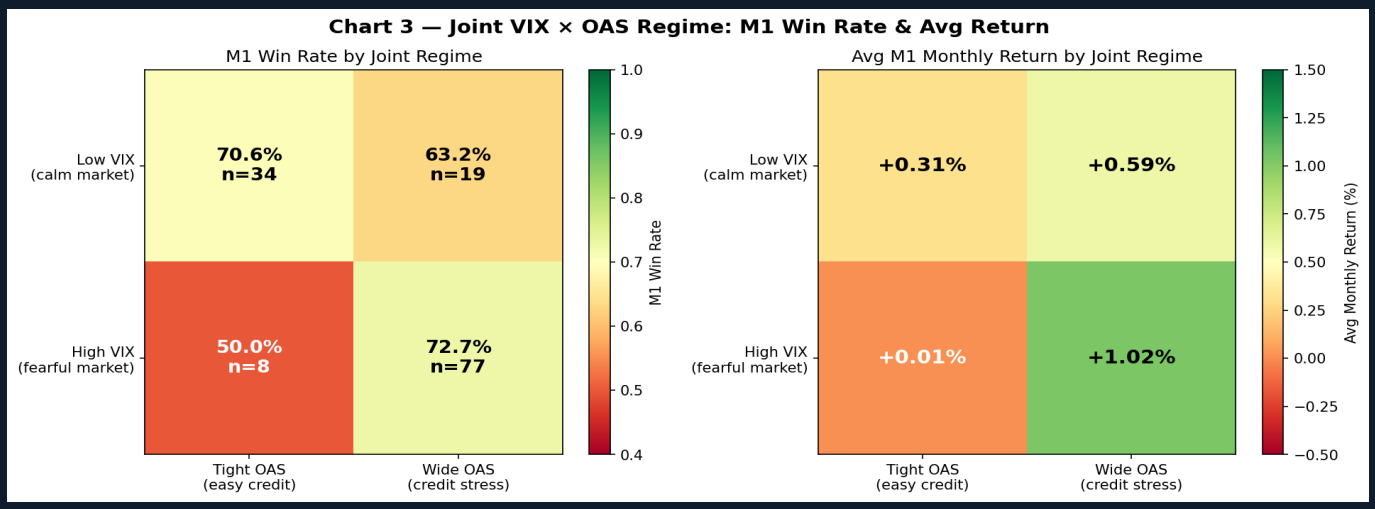


Tight OAS: 66.7% | Wide OAS: 70.8% | Chi-square p = 0.7730 (not significant)

Interpretation

OAS tells the same story as VIX. Wide spreads (stressed credit) are not associated with worse M1 performance. p = 0.77 is far from significance.

Chart 3 - The most striking result: combined VIX + OAS regimes



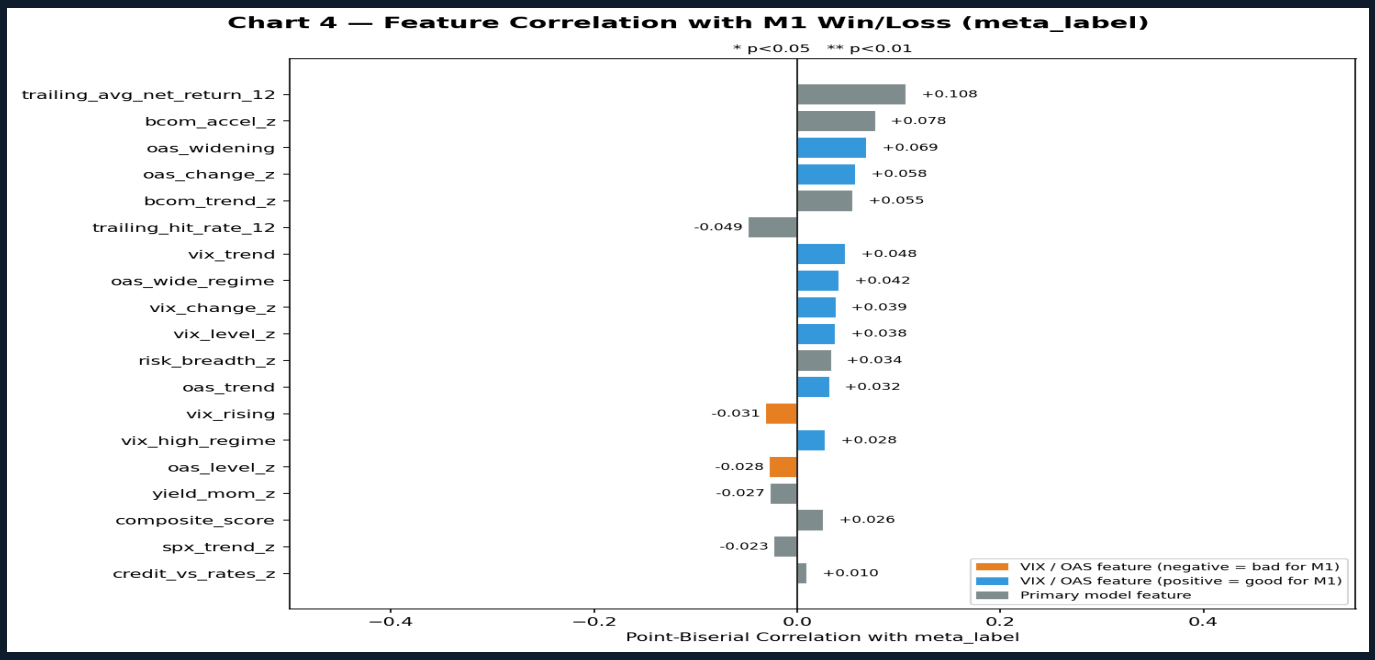
High VIX + Wide OAS (worst expected) delivered 72.7% win rate and +1.02% avg return - the best of all four regimes.

The counter-intuitive finding

The regime most investors would label 'dangerous' produced the best M1 performance. This strongly supports the thesis that M1's SELL logic already capitalises on stress periods by positioning defensively.

5 continued - Feature Correlations and Regime Simulation

Chart 4 - Feature correlations with M1 win/loss



*No feature exceeds 0.11 correlation with meta_label. None are statistically significant (no * markers).*

What this means

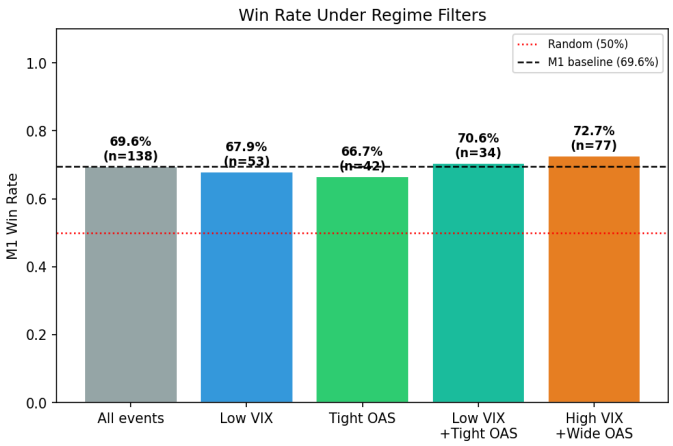
The highest correlation with M1 winning is only 0.108 (trailing average return). All 10 VIX and OAS features are below 0.07. In a dataset of 138 observations, you would need at least 0.17 correlation to be statistically significant at $p < 0.05$. None of our features come close.

Chart 6 - What if we used regime as a simple gate?

Chart 6 — Simulated M2 Regime Gate: If You Only Traded in Favorable Regimes

Performance Under Different Regime Filters

Filter	N events	Win Rate	Avg Return	Sharpe	Max Loss
All M1 events (no filter)	138	69.6%	+0.73%	1.19	-5.39%
Low VIX only	53	67.9%	+0.41%	0.84	-3.32%
Tight OAS only	42	66.7%	+0.25%	0.55	-3.32%
Low VIX + Tight OAS (best regime)	34	70.6%	+0.31%	0.68	-3.32%
High VIX + Wide OAS (worst regime)	77	72.7%	+1.02%	1.49	-5.39%



Filtering by 'favourable' VIX+OAS reduces Sharpe from 1.19 to 0.68. The 'bad' regime (High VIX+Wide OAS) has Sharpe 1.49.

SECTION 6

CONCLUSIONS & NEXT STEPS

What we learned and where we go from here

This iteration of M2 did not add value. But the framework, the data pipeline, and the walk-forward infrastructure are all now in place. We have a clear diagnostic of why this approach failed, which directly informs the next steps.

6 - Summary and Next Steps

Honest findings and a roadmap for improvement

What we built

In this research sprint we built an end-to-end meta-labeling system: a 28-year dataset of 138 labeled M1 signals, 22 features including 10 new VIX and OAS regime features, a leakage-free walk-forward training framework, and a logistic regression M2 classifier. The entire codebase is modular, tested, and version-controlled.

What we found

- **M1 is strong on its own**
  71.8% win rate and 1.31 annualised Sharpe over 28 years. Outperforms 60/40 and simple buy-and-hold on a risk-adjusted basis.
- **M2 did not improve M1**
  ROC-AUC of 0.523 (barely above random). M2-approved trades had slightly lower Sharpe (1.18) than simply trading all M1 signals (1.31).
- **VIX and OAS carry no independent signal**
  Chi-square tests (p=0.89, p=0.77) confirm no statistically significant relationship between these regime features and M1 win/loss outcomes.
- **M1 is regime-robust - a feature, not a bug**
  M1 performs equally well or better in stressed markets, likely because its SELL logic correctly positions defensively during downturns.
- **Sample size is a binding constraint**
  138 monthly events over 28 years is too small to detect subtle regime effects even if they exist. This is the most honest structural limitation.

What comes next

01	**Find features orthogonal to M1** The key lesson is that VIX and OAS are redundant with M1's existing inputs. The next iteration should look for information M1 structurally cannot see: sentiment data, options market signals, cross-asset dispersion.
02	**Investigate ROC curve threshold optimisation** The 0.5 threshold was never optimised. A walk-forward threshold selection within the training folds may unlock value that fixed thresholds miss.
03	**Cross-validation across multiple window sizes** Test minimum training windows of 3, 6, 12, 24, and 36 months to understand how much history M2 actually needs and whether results are stable.
04	**Separate BUY and SELL meta-labelers** BUY and SELL signals may have different risk profiles. Training one M2 for each direction could improve precision at the cost of sample size.
05	**Production integration once validated** Once a specification shows genuine out-of-sample improvement, integrate it into the live monthly allocation process with defined guardrails.

A null result is a real result. We now know that VIX and OAS do not help M2, that M1 is robust across market regimes, and that the binding constraint is sample size and feature orthogonality. This diagnostic is what makes the next iteration

faster and more targeted. The infrastructure is built. The next question is sharper.